

Adversarial Uncertainty Injection: A Framework for Designing Robust Rule-Based Expert Systems Against Incomplete Inputs

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Abstract

Rule-based expert systems can perform expert-level reasoning in specific domains, but remain vulnerable to incomplete or missing data that frequently occur in real-world environments. Existing approaches treat incomplete data as unknown values and proceed with approximate reasoning (Quinlan, 1989), which leads to instability in certainty factor (CF) values and degrades the reliability of final conclusions. In this paper, we propose the Adversarial Uncertainty Injection (AUI) framework to address this problem. AUI deliberately injects three types of uncertainty (missing values, CF distortion, and contradiction) during the rule design phase, enabling experts to simulate real-world incomplete data scenarios in advance and design robust rule sets and CF values accordingly.

1. Introduction

Rule-based expert systems have demonstrated expert-level performance across diverse domains since the 1980s, including medical diagnosis, mineral exploration, and military decision-making. Systems such as MYCIN, PROSPECTOR, and DENDRAL have leveraged IF-THEN rules and certainty factor (CF) theory to derive reliable conclusions within constrained problem domains.

However, one of the most significant challenges facing expert systems in real-world deployment is **data incompleteness**. In practice, input data is rarely complete due to sensor failures, measurement omissions, and inconsistencies in expert judgment. The challenge of handling unknown attribute values has been a persistent concern in the literature (Quinlan, 1989), and similar issues have recently been explored in the context of graph learning with missing features (Um et al., 2025). Nevertheless, these approaches share a common limitation: they are fundamentally *reactive*, addressing incomplete data only after it has occurred.

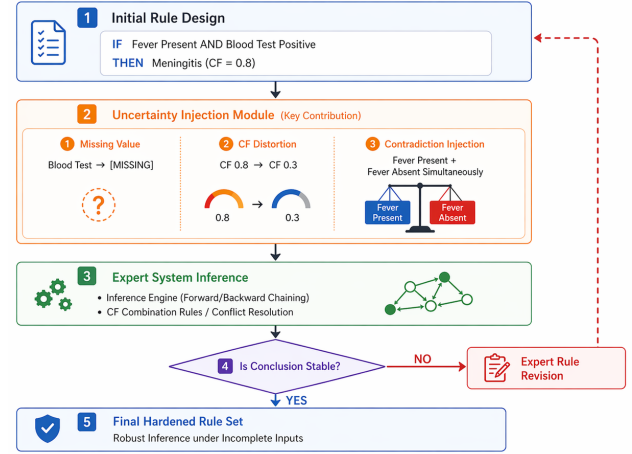


Figure 1. Overview of Adversarial Uncertainty Injection (AUI). AUI deliberately injects three types of uncertainty (missing value, CF distortion, and contradiction) during the rule design phase, enabling experts to simulate incomplete scenarios in advance and iteratively harden the rule set against real-world incomplete inputs.

From the perspective of expert system validation, (O’Keefe et al., 1987) has long emphasized that robustness under incomplete inputs is one of the most critical yet underaddressed criteria for high-stakes systems such as those deployed in medical or military contexts. Despite this, no systematic methodology has been proposed for designing expert systems that are inherently robust to incomplete inputs from the outset.

To address this fundamental gap, we propose **Adversarial Uncertainty Injection (AUI)**, a framework that deliberately injects uncertainty during the rule design phase, enabling experts to simulate diverse incomplete scenarios and derive robust rules and CF values proactively.

In summary, our contributions are as follows:

- We reframe the problem of incomplete data in rule-based expert systems from a **proactive** design perspective.
- We propose three uncertainty injection mechanisms: **missing value injection**, **CF distortion injection**, and **contradiction injection**.

- We present a systematic framework that integrates with existing CF theory and conflict resolution strategies to support robust rule design.

Problem Definition and Limitations

Uncertainty in expert systems arises from weak IF-THEN correlations, ambiguous natural language, missing data (Quinlan, 1989), and conflicting expert opinions.

Bayesian inference quantifies uncertainty using prior and conditional probabilities. However, it requires reliable statistical data, assumes conditional independence among evidence, and becomes computationally intractable as the knowledge base scales.

Certainty factor (CF) theory addresses uncertainty by assigning numerical confidence values based on expert judgment. However, CF values are inherently subjective, and the CF combination formulas can yield unreliable results when input data is severely incomplete.

Both approaches are fundamentally reactive — they respond to incomplete data after it arises, rather than designing systems that are robust to such situations from the start. This robustness gap has been identified as a critical practical limitation in expert system validation research (O’Keefe et al., 1987).

Proposed Method: Adversarial Uncertainty Injection (AUI)

The core philosophy of AUI is: “**train the system to withstand incomplete situations before they occur.**” By deliberately injecting uncertainty during the rule design phase, AUI enables experts to simulate diverse incomplete scenarios and derive robust rules and CF values proactively. This notion of deliberate injection is conceptually aligned with recent work on missing feature imputation (Um et al., 2025), but AUI repurposes it as an *adversarial design tool* for stress-testing IF-THEN rule robustness.

AUI consists of three stages.

Stage 1: Initial Rule Design. A knowledge engineer designs a complete IF-THEN rule set with CF values.

Stage 2: Uncertainty Injection. Starting from the complete rule set, three types of uncertainty are injected:

- **Missing Value Injection:** Removes part of the antecedent (IF condition) data to simulate incomplete input scenarios. For example, omitting blood test results in a medical diagnosis system.
- **CF Distortion Injection:** Deliberately distorts CF values to simulate uncertainty in expert judgment. For example, reducing a CF value from 0.9 to 0.4.

- **Contradiction Injection:** Simultaneously inputs conflicting evidence to simulate contradictory scenarios. For example, injecting both “fever present” and “fever absent” at the same time.

Stage 3: Expert Review and Rule Hardening. The expert reviews inference results produced under the injected uncertainty conditions. When conclusions are found to be unstable, the expert directly revises the rules and CF values to harden them. Repeating this cycle produces a rule set that is robust to incomplete real-world inputs.

AUI requires no structural modification: conflict resolution strategies handle contradictions, CF combination formulas are tuned for stability under distortion, and backward chaining re-establishes sub-goals under missing antecedents.

Feasibility and Limitations

AUI can be implemented directly on top of existing expert system shells without modifying the underlying inference engine or CF computation structure. The uncertainty injection component operates as a separate simulation module, immediately applicable to high-stakes domains such as medical diagnosis and military decision-making (O’Keefe et al., 1987).

Limitation 1: There is no clear criterion for determining the type and intensity of uncertainty to inject. **Mitigation:** Bayesian inference can be used to estimate the probability distribution of missing data patterns in each domain, providing a principled basis for determining injection intensity.

Limitation 2: The expert review burden increases. **Mitigation:** Meta-rules can automatically pass stable rules and flag only unstable ones for review.

Limitation 3: Injection scenarios may not cover all real-world cases. **Mitigation:** Missing patterns from actual operations can continuously update the injection scenario pool.

References

- O’Keefe, R. M., Balci, O., and Smith, E. P. Validating expert system performance. *IEEE Expert*, 2(4):81–90, 1987. doi: 10.1109/MEX.1987.5006538.
- Quinlan, J. R. Unknown attribute values in induction. In *Proceedings of the sixth international workshop on machine learning*, pp. 164–168. Elsevier, 1989.
- Um, D., Kim, S., Park, J., Lim, J., Ahn, S. J., and Park, S. Propagate and inject: Revisiting propagation-based feature imputation for graphs with partially observed features. In *Forty-second International Conference on Machine Learning*, 2025.